

Brownian Motion and Itô Calculus

Financial Engineering · Section 3

Zong-Wei Yeh

Department of Finance, National Chung Cheng University

Fall 2026

Outline

- 3.1 From random walk to Brownian motion
- 3.2 The Itô integral
- 3.3 Itô processes and stochastic differential equations
- 3.4 Itô's lemma
- 3.5 Covariation, the product rule and the quotient rule
- 3.6 Mean reversion: the Ornstein–Uhlenbeck process
- 3.7 Change of measure: Radon–Nikodym and Girsanov
- 3.8 The Feynman–Kac theorem
- 3.9 Martingale representation

From Random Walk to Brownian Motion

What the limit theorem does and does not give

- Scale the walk by $\sqrt{\Delta t}$, so $\text{Var}(W_t^{(n)}) = n\Delta t = t$ for every n .
- Steps of size Δt give a limit that is identically zero; steps of size $(\Delta t)^{1/3}$ explode. Only the exponent $\frac{1}{2}$ works.
- The CLT then fixes the limit law at any *finite* list of dates. That is all it does.
- Convergence of the whole path, and continuity of the limit, is **Donsker's invariance principle**, which we import.

Brownian motion

Let $(\mathcal{F}_t)$ be the filtration generated by the path of the process below, so that date- t information is the trajectory observed so far.

Definition 3.1 (Brownian motion)

$W = (W_t)_{t \geq 0}$ is a *standard Brownian motion* if

- (i) $W_0 = 0$ and the paths $t \mapsto W_t$ are continuous;
- (ii) for $s < t$, $W_t - W_s$ is independent of $\mathcal{F}_s$;
- (iii) $W_t - W_s \sim N(0, t - s)$.

— Existence is a real theorem, which we import.

The whole law, not just one date

Proposition 3.2 (Joint density of Brownian motion)

For $0 = t_0 < t_1 < \dots < t_n$ and $\Delta_k = t_k - t_{k-1}$, the vector $(W_{t_1}, \dots, W_{t_n})$ has density

$$f(x_1, \dots, x_n) = \prod_{k=1}^n \frac{1}{\sqrt{2\pi\Delta_k}} \exp\left(-\frac{(x_k - x_{k-1})^2}{2\Delta_k}\right), \quad x_0 = 0.$$

- One factor per time step, each depending only on how far the path moved during that step. That is the independent-increment property written as a formula.
- Every conditional calculation below is read off this density.

Example 3.3

Example 3.3 (Conditional moments)

Let W be a standard Brownian motion and $t < s$. Show that

$$\begin{aligned} (1) \mathbb{E}[W_s \mid W_t = B] &= B, & (2) \operatorname{Var}(W_s \mid W_t = B) &= s - t, \\ (3) \mathbb{E}[W_t \mid W_s = B] &= B \frac{t}{s}, & (4) \operatorname{Var}(W_t \mid W_s = B) &= \frac{(s - t)t}{s}, & (5) \mathbb{E}[W_t W_s] &= t. \end{aligned}$$

Example 3.4

Example 3.4 (Transformations that preserve the law)

Show that each of the following is again a standard Brownian motion:

$$(1) X_t = a W_{t/a^2} \ (a > 0), \quad (2) Y_t = t W_{1/t} \ (t > 0), \ Y_0 = 0,$$

$$(3) Z_u = W_{t+u} - W_t \quad (t \text{ fixed}).$$

Quadratic variation

Theorem 3.5 (Quadratic variation)

Partition $[0, t]$ into n equal steps. Then

$$Q_n = \sum_{i=1}^n (W_{t_i} - W_{t_{i-1}})^2 \longrightarrow t$$

in mean square, hence in probability, written $[W]_t = t$, or $(dW_t)^2 = dt$.

- The sum of squared shocks over any horizon is *not random* in the limit. It equals the length of the horizon.
- That is why the chain rule of §3.4 carries an extra non-random term.



The Itô Integral

Why we need this before anything else

- Every model in this course will be written as $dS_t = \mu S_t dt + \sigma S_t dW_t$, or something like it.
- The dt part is an ordinary integral. The dW part is not, and so far nothing says what $\int \sigma dW$ is.
- Same integral, second reading: if θ_s is the position held and dW_s the shock, $\int_0^T \theta_s dW_s$ is the trading gain. Building the integral and building the gains process are one job.
- Riemann–Stieltjes will not do it: $\sum_i |\Delta W_i| \approx t/\sqrt{\Delta t} \rightarrow \infty$, so Brownian paths have infinite variation.

The Itô integral

Definition 3.6 (Itô integral of a simple integrand)

For θ *simple*, that is constant at the adapted value $\theta_{t_{i-1}}$ on each interval $(t_{i-1}, t_i]$ of a partition of $[0, T]$,

$$I_T = \int_0^T \theta_s dW_s = \sum_{i=1}^n \theta_{t_{i-1}} (W_{t_i} - W_{t_{i-1}}).$$

We write I_t for the same sum up to t , with the last holding period cut at t .

- Position held, times shock received, summed over holding periods.

Why the left endpoint

- The integrand sits at the **left** endpoint: the position is chosen before the shock arrives, as on the tree.
- Left and right endpoints differ by $Q_n \rightarrow t \neq 0$, so this is a modeling decision, not a technicality.

What the integral inherits

Theorem 3.7 (Martingale property and Itô isometry)

For simple adapted θ with $\mathbb{E} \int_0^T \theta_s^2 ds < \infty$:

- (a) I_t is a martingale; in particular $\mathbb{E}[I_T] = 0$;
 - (b) $\mathbb{E}[I_T^2] = \mathbb{E} \int_0^T \theta_s^2 ds$.
- (a) Trading against a fair shock earns nothing on average.
 - (b) The *risk* of the gains is the accumulated squared position. The isometry is the risk accounting of continuous trading.
 - Part (a) is what will prove Proposition 3.11 in §3.3.

Itô Processes and Stochastic Differential Equations

The language every model is written in

Definition 3.8 (Itô process)

For adapted μ, σ with $\int_0^t |\mu_s| ds < \infty$ and $\int_0^t \sigma_s^2 ds < \infty$ for every t , almost surely, the process X is an *Itô process* if

$$X_t = X_0 + \int_0^t \mu_s ds + \int_0^t \sigma_s dW_s \quad \text{for all } t, \quad \text{written} \quad dX_t = \mu_t dt + \sigma_t dW_t.$$

We call μ the *drift* and σ the *volatility* of X .

- The integral equation is the definition; the differential form abbreviates it. There is no dW_t/dt to write down.
- The second integral is the Itô integral of §3.2, which is why that subsection had to come first.

When the coefficients depend on the state

Definition 3.9 (Stochastic differential equation)

A *stochastic differential equation* is the case where the drift and the volatility are functions of time and of the current state,

$$dX_t = \mu(t, X_t) dt + \sigma(t, X_t) dW_t, \quad X_0 \text{ given,}$$

and X *solves* it if it satisfies the corresponding integral equation for all t .

- So an SDE is a recipe. The drift and the volatility are read off the date and the state the process is in right now.

When an SDE has a solution

- Existence and uniqueness hold when μ and σ are Lipschitz in the *state*, uniformly in t , and grow at most linearly in it. Measurability in t is enough. We import this.

Example 3.10

Example 3.10 (The three models of this course)

- (a) **Money account.** $dB_t = rB_t dt$: $\mu(t, x) = rx$, $\sigma \equiv 0$, so $B_t = B_0 e^{rt}$.
 - (b) **Geometric Brownian motion.** $dS_t = \mu S_t dt + \sigma S_t dW_t$: both coefficients proportional to the state, so shocks are *percentages*. Solved in §3.4.
 - (c) **Ornstein–Uhlenbeck.** $dX_t = \kappa(\bar{\theta} - X_t) dt + \sigma dW_t$, $\kappa > 0$: the drift changes sign at $\bar{\theta}$. Solved in §3.6.
- The three differ only in how μ and σ depend on the state. That choice is the whole modeling decision.

Martingale means no drift

Proposition 3.11 (Martingale $\iff$ no drift)

Let $dX_t = \mu_t dt + \sigma_t dW_t$ on $[0, T]$ with μ, σ adapted, $\mathbb{E}|X_0| < \infty$, $\mathbb{E} \int_0^T |\mu_s| ds < \infty$ and $\mathbb{E} \int_0^T \sigma_s^2 ds < \infty$. Then X is a martingale **if and only if** $\mu_t = 0$ for almost every t , almost surely.

- Used in both directions: martingale $\Rightarrow$ zero drift is the Feynman–Kac step of §3.8; zero drift $\Rightarrow$ martingale is the hedging argument of Section 4.
- Without $\mathbb{E} \int_0^T \sigma_s^2 ds < \infty$ a driftless process is in general only a *local* martingale. A positive one is a supermartingale, so its expected value can fall. That is how bubbles are modeled.

Itô's Lemma

Itô's lemma

Theorem 3.12 (Itô's lemma)

Let $dX_t = \mu_t dt + \sigma_t dW_t$ and let $f(t, x)$ have continuous f_t, f_x, f_{xx} . Then

$$df = \left(f_t + \mu_t f_x + \frac{1}{2} \sigma_t^2 f_{xx} \right) dt + \sigma_t f_x dW_t.$$

Mnemonicly: expand to second order and use $(dW)^2 = dt$, $dW dt = 0$, $(dt)^2 = 0$.

- A function of X inherits the shock in X . Its drift has three sources: time passing, the drift of X , and curvature turning volatility into drift.
- That last term $\frac{1}{2} \sigma^2 f_{xx}$ is positive wherever $\sigma_t \neq 0$ and $f_{xx} > 0$. Convexity raises an expectation, which is Jensen's inequality. That is how volatility enters an option price.

Example 3.13

Example 3.13 (Geometric Brownian motion)

Solve

$$dS_t = \mu S_t dt + \sigma S_t dW_t, \quad S_0 > 0,$$

by applying Itô's lemma to $\ln S_t$.

Covariation, the Product Rule and the Quotient Rule

How two paths move together

Definition 3.14 (Quadratic covariation)

For Itô processes X, Y , partition $[0, t]$ into n equal steps. If

$$C_n = \sum_{i=1}^n (X_{t_i} - X_{t_{i-1}})(Y_{t_i} - Y_{t_{i-1}})$$

converges in probability, the limit is $[X, Y]_t$; taking $Y = X$ gives $[X]_t$.

- A covariance averages across states of the world. A covariation is built from the increments of the two paths, with no such averaging.
- That is why a change of measure cannot touch it (§3.7).

Correlated Brownian motions

Proposition 3.15 (Correlated Brownian motions)

Let $W^1, W^\perp$ be independent standard Brownian motions, fix $\rho \in [-1, 1]$, and set $W_t^2 = \rho W_t^1 + \sqrt{1 - \rho^2} W_t^\perp$. Then W^2 is a standard Brownian motion, $\text{Corr}(W_t^1, W_t^2) = \rho$ for $t > 0$, and

$$[W^1, W^2]_t = \rho t, \quad \text{i.e.} \quad dW_t^1 dW_t^2 = \rho dt.$$

- One number ρ describes how the two shocks move together. It does not depend on the horizon or on the partition.
- So a two-factor model needs two volatilities and one correlation. Section 7's Heston model is exactly that.

Itô's lemma in two variables

Theorem 3.16 (Two-variable Itô formula)

With $dX = \mu^X dt + \sigma^X dW^1$, $dY = \mu^Y dt + \sigma^Y dW^2$, $dW^1 dW^2 = \rho dt$, and $f(t, x, y)$ having continuous $f_t, f_x, f_y, f_{xx}, f_{xy}, f_{yy}$,

$$df = \left(f_t + \mu^X f_x + \mu^Y f_y + \frac{1}{2}(\sigma^X)^2 f_{xx} + \rho \sigma^X \sigma^Y f_{xy} + \frac{1}{2}(\sigma^Y)^2 f_{yy} \right) dt \\ + \sigma^X f_x dW^1 + \sigma^Y f_y dW^2.$$

- One new term: cross-curvature times the covariation rate. No factor $\frac{1}{2}$, because the mixed partial appears twice in the Hessian.

The product rule

Corollary 3.17 (Itô product rule)

Under the assumptions of Theorem 3.16,

$$d(X_t Y_t) = X_t dY_t + Y_t dX_t + d[X, Y]_t, \quad d[X, Y]_t = \rho \sigma_t^X \sigma_t^Y dt.$$

- Ordinary calculus gives the first two terms. The third is the Itô correction, and it is the covariation.
- If either process has no dW term, the correction is zero.
- Taking $Y = X$ gives $[X]_t = \int_0^t (\sigma_s^X)^2 ds$, the general form of Theorem 3.5.

The quotient rule

Corollary 3.18 (Itô quotient rule)

Under the assumptions of Theorem 3.16, on any interval where $Y_t \neq 0$,

$$d\left(\frac{X_t}{Y_t}\right) = \frac{dX_t}{Y_t} - \frac{X_t}{Y_t^2} dY_t - \frac{d[X, Y]_t}{Y_t^2} + \frac{X_t}{Y_t^3} d[Y]_t.$$

- The first two terms are the ordinary quotient rule. The two Itô corrections enter with opposite signs.

Reading the two corrections

- Positive covariation between numerator and denominator lowers the drift of the ratio.
- Volatility in the denominator alone raises it when $X_t > 0$ and $Y_t > 0$, because x/y is then convex in y .
- The two worked cases that follow both have a positive numerator and a positive denominator.

Example 3.19

Example 3.19 (The discounted price)

Let $dS_t = \mu S_t dt + \sigma S_t dW_t$ with $S_0 > 0$, and let $B_t = e^{rt}$ be the money account of Example 3.10, with $B_0 = 1$ and r constant. Find the dynamics of the discounted price $\tilde{S}_t = S_t/B_t$, and say why both Itô corrections of Corollary 3.18 drop out.

Example 3.20

Example 3.20 (Ratio of two geometric Brownian motions)

Let $dS_t^i = \mu_i S_t^i dt + \sigma_i S_t^i dW_t^i$ for $i = 1, 2$, with $S_0^1, S_0^2 > 0$ and $\text{Corr}(W^1, W^2) = \rho$. Find the dynamics of $R_t = S_t^1 / S_t^2$.

Mean Reversion: the Ornstein–Uhlenbeck Process

Mean reversion

Proposition 3.21 (The Ornstein–Uhlenbeck process)

$dX_t = \kappa(\bar{\theta} - X_t)dt + \sigma dW_t$, $\kappa > 0$, is solved by

$$X_t = \bar{\theta} + (X_0 - \bar{\theta})e^{-\kappa t} + \sigma \int_0^t e^{-\kappa(t-s)} dW_s,$$

which is normal with $\mathbb{E}[X_t] = \bar{\theta} + (X_0 - \bar{\theta})e^{-\kappa t}$ and $\text{Var}(X_t) = \frac{\sigma^2}{2\kappa}(1 - e^{-2\kappa t}) \rightarrow \frac{\sigma^2}{2\kappa}$.

- The effect of the starting point decays at exponential rate κ , half-life $\ln 2/\kappa$.
- The process settles into a stationary normal law around $\bar{\theta}$, with variance $\sigma^2/(2\kappa)$. Brownian motion has $\text{Var} = t \rightarrow \infty$; mean reversion caps it.

Example 3.22

Example 3.22

A variance process is OU with $\kappa = 2$, $\bar{\theta} = 0.04$, $\sigma = 0.2$, starting at $X_0 = 0.10$ after a market stress. Find $\mathbb{E}[X_t]$ and $\text{Var}(X_t)$ at $t = 0.5$, and the stationary distribution.

Mean reversion is not an arbitrage

- An OU process is normal, so it can go negative. Fine for a spread, not for a variance. The Cox–Ingersoll–Ross (CIR) model replaces σ by $\sigma\sqrt{X}$.
- The common argument against reverting *prices* is wrong. Predictable mean reversion under $\mathbb{P}$ is not by itself an arbitrage: the strategy still carries risk and can lose money.
- What no-arbitrage restricts is different. Discounted prices must be martingales under an equivalent martingale measure $\mathbb{Q}$, which is Section 2's fundamental theorem.
- Drift under $\mathbb{P}$ is a risk premium, and §3.7 moves it. Drift under $\mathbb{Q}$ is not free: for a traded asset paying no dividend it equals r .

Change of Measure: Radon–Nikodym and Girsanov

The Radon–Nikodym derivative

Definition 3.23 (Radon–Nikodym derivative)

Let $\mathbb{P}$ and $\mathbb{Q}$ be equivalent measures on a finite Ω , with state probabilities π_ω under $\mathbb{P}$ and q_ω under $\mathbb{Q}$; equivalent means $\pi_\omega > 0 \iff q_\omega > 0$. Then

$$Z(\omega) = \frac{d\mathbb{Q}}{d\mathbb{P}}(\omega) = \frac{q_\omega}{\pi_\omega} > 0, \quad \mathbb{E}^{\mathbb{Q}}[X] = \mathbb{E}^{\mathbb{P}}[ZX] \text{ for every } X, \quad \mathbb{E}^{\mathbb{P}}[Z] = 1.$$

- You met Z in Section 1 without its name: there $q_s/\pi_s = (1+r)m_s$. The stochastic discount factor is, up to discounting, exactly this object.

What is proved for a finite Ω

- The two identities are one-line sums when Ω is finite.
- On a general space the existence of Z is the Radon–Nikodym theorem. The abstract Bayes rule that follows holds in the same form. We import both.
- The formulas do not change.

Forecasting under the new measure

Write $Z_t = \mathbb{E}^{\mathbb{P}}[Z \mid \mathcal{F}_t]$ for the *density process*: a positive $\mathbb{P}$ -martingale with $Z_0 = 1$.

Proposition 3.24 (Abstract Bayes rule)

Let $\mathbb{E}^Q|X| < \infty$. Then at every date t ,

$$\mathbb{E}^Q[X \mid \mathcal{F}_t] = \frac{\mathbb{E}^{\mathbb{P}}[ZX \mid \mathcal{F}_t]}{Z_t}.$$

At $t = 0$ this is Definition 3.23's identity, since $\mathcal{F}_0$ is trivial and $Z_0 = 1$.

- Reweight by Z , then renormalize. The denominator corrects the reweighting that has already happened by date t .

The density process

Lemma 3.25 (The density process is the exponential martingale)

Fix a constant λ and let $Z_T = e^{-\lambda W_T - \lambda^2 T/2}$. Then

$$Z_t = \mathbb{E}^{\mathbb{P}}[Z_T \mid \mathcal{F}_t] = e^{-\lambda W_t - \lambda^2 t/2}, \quad 0 \leq t \leq T.$$

- $e^{-\lambda W_t - \lambda^2 t/2}$ is the *exponential martingale*. Itô's lemma makes its drift zero.
- Its volatility is square integrable on any bounded horizon, so Proposition 3.11 makes it a martingale with mean 1.

Using the density process

- The reweighting accumulates multiplicatively through time. So changing the measure at T and changing it along the way are the same operation.
- This lets Section 4 treat the risk-neutral measure as a running object rather than a terminal-date trick.

Girsanov

Theorem 3.26 (Girsanov)

For a constant λ , define Q by $\frac{dQ}{d\mathbb{P}} = Z_T = e^{-\lambda W_T - \lambda^2 T/2}$. Then under Q the process $\tilde{W}_t = W_t + \lambda t$ is a standard Brownian motion on $[0, T]$.

- A change of measure can add or remove **drift**, and that is *all* it can do. Here λ is Section 4's *market price of risk*.
- For an adapted (λ_t) take $Z_T = \exp(-\int_0^T \lambda_s dW_s - \frac{1}{2} \int_0^T \lambda_s^2 ds)$, and assume Novikov: $\mathbb{E}[\exp(\frac{1}{2} \int_0^T \lambda_s^2 ds)] < \infty$. It makes Z a true martingale, so $\mathbb{E}^{\mathbb{P}}[Z_T] = 1$. That general case is imported.

Why volatility survives the change of measure

- Quadratic variation is computed path by path. Theorem 3.5 gives convergence in probability; along a refining sequence of partitions it is almost sure, which we import.
- So $[W]_t = t$ on a set of $\mathbb{P}$ -probability one.
- Equivalent measures share their null events, so that set has Q -probability one too.
- Two investors may disagree about probabilities, and so about drift. They cannot disagree about realized variance.
- Note the wording: equivalent measures agree on which events are *null*, not on the probability of every event.



The Feynman–Kac Theorem

Feynman–Kac: the setup

Throughout: X solves $dX_t = \mu(t, X_t) dt + \sigma(t, X_t) dW_t$, the rate r is constant, h is a payoff function, and $F(t, x) = \mathbb{E}[e^{-r(T-t)} h(X_T) \mid X_t = x]$.

Definition 3.27 (Reached points)

Let X be an Itô process. A point (t, x) is *reached* when every interval around x carries positive probability under the law of X_t : $\mathbb{P}(|X_t - x| < \delta) > 0$ for every $\delta > 0$. The set of such x is the *support* of the law of X_t .

- Reached points are the only place $F(t, x)$ says anything, since it conditions on $X_t = x$.

What Feynman–Kac needs

- h is continuous, with $\mathbb{E}[|h(X_T)|] < \infty$.
- μ and σ are continuous in (t, x) . §3.3's Lipschitz import gives continuity in x only.
- F is continuous on $[0, T] \times \mathbb{R}$, with F_t, F_x, F_{xx} continuous on $[0, T) \times \mathbb{R}$. A call payoff kinks at the strike, so F_x cannot be continuous up to T . Where σ is not zero, the expectation before T smooths it.
- Proposition 3.11's two conditions on the drift and the volatility, up to a bounded factor, applied to $e^{-rt}F(t, X_t)$.

Feynman–Kac

Theorem 3.28 (Feynman–Kac)

Under the assumptions just listed, with X , r , h and F as on the setup frame,

$$F_t + \mu F_x + \frac{1}{2}\sigma^2 F_{xx} = rF$$

at every reached (t, x) with $t < T$, and $F(T, x) = h(x)$ at every reached (T, x) .

Conversely, a G with the same regularity, solving the same equation at those points with the same terminal values and the same condition on G_x , equals that expectation there.

- The expectation and the PDE are two readings of one object.

Reading Feynman–Kac

- A *diffusion* is a solution of an SDE of §3.3. A discounted conditional expectation of one solves a PDE wherever the process goes. Its coefficients are the drift of X and half its squared volatility.
- And any well-behaved solution of that PDE is such an expectation. Pricing by simulation and pricing by solving an equation give the same number. Section 5 does both.
- Going from X_t to $\mathcal{F}_t$ needs the Markov property.

Martingale Representation

Martingale representation

Theorem 3.29 (Martingale representation)

For the filtration generated by W , every square-integrable martingale M can be written

$$M_t = M_0 + \int_0^t \phi_s dW_s$$

for some adapted ϕ .

- In a world whose only news source is W , every square-integrable fair game is a trading gain.

Why the hedge exists

- This is the continuous-time counterpart of binomial completeness (Section 5 proves it): one source of risk, traded by one asset, spans everything.
- Girsanov, Feynman–Kac and this theorem are the three moves Section 4 assembles. We import this one rather than prove it.